🧪 Lab 2: Secure & Scalable VLAN Design (Router-on-a-Stick)

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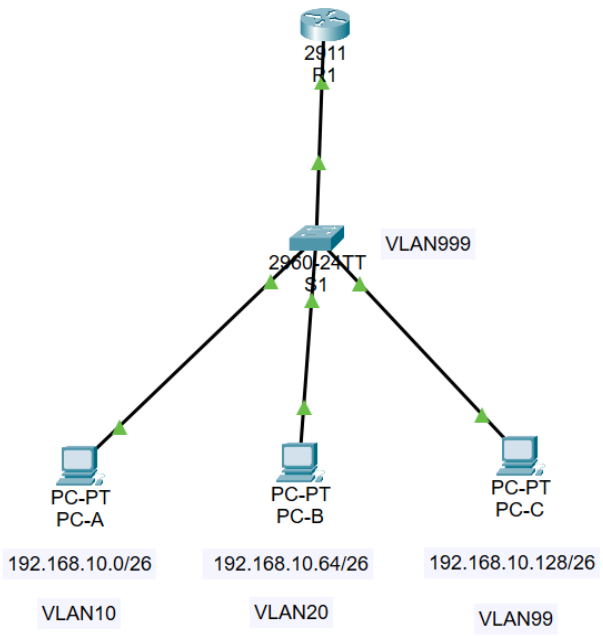
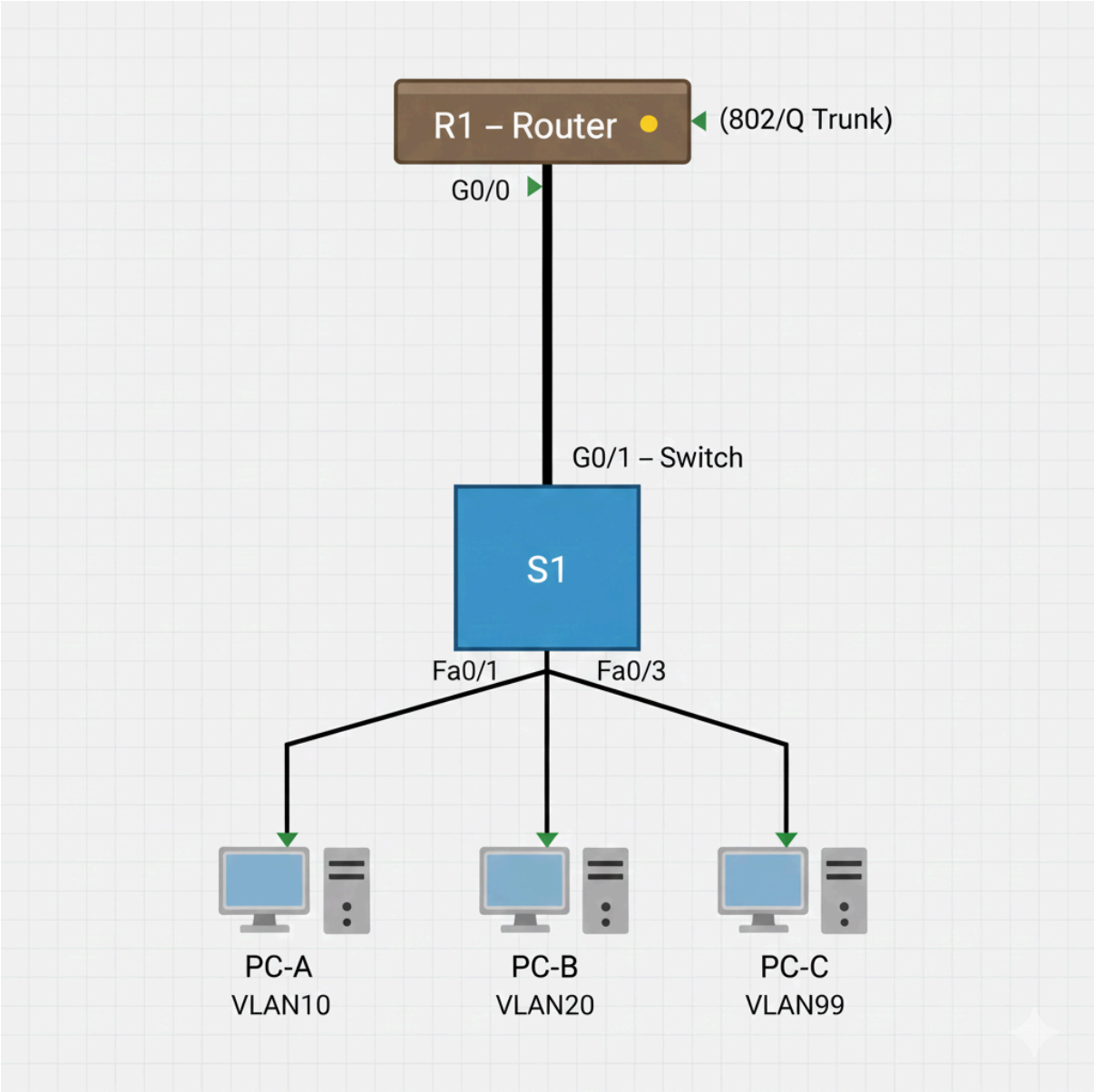
📊 TOPOLOGY DIAGRAM

None

```
                    [ R1 – Router ]
                    G0/0 (802.1Q Trunk)
                              |
                              |
                     [ G0/1 – Switch ]
                    ┌─────────┴─────────┐
                    |        S1         |
                    |         |         |
              [Fa0/1]      [Fa0/2]     [Fa0/3]
                  |         |           |
                  |         |           |
                [PC-A] [PC-B]          [PC-C]
                VLAN10  VLAN20         VLAN99
```





📋 ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
- **Subnet Mask (/26):** 255.255.255.192
- **Subnets Used:**
  - VLAN 10 → 192.168.10.0/26
  - VLAN 20 → 192.168.10.64/26
  - VLAN 99 → 192.168.10.128/26

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

🎯 LAB OBJECTIVES

By completing this lab, you will be able to:

Part 1: Design and configure multiple VLANs on a switch **Part 2:** Implement Router-on-a-Stick for inter-VLAN routing **Part 3:** Apply subnetting using /26 networks **Part 4:** Isolate management and unused ports for security **Part 5:** Verify and troubleshoot Layer 2 and Layer 3 connectivity

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🔧 LAB INSTRUCTION STEPS

PART 1: SWITCH CONFIGURATION (35 minutes)

Step 1: Basic Switch Setup

```
None
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
```

Step 2: Create VLANs

```
None
S1(config)# vlan 10
S1(config-vlan)# name USERS
S1(config-vlan)# exit

S1(config)# vlan 20
S1(config-vlan)# name SERVERS
S1(config-vlan)# exit

S1(config)# vlan 99
S1(config-vlan)# name MANAGEMENT
S1(config-vlan)# exit

S1(config)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config-vlan)# exit
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname S1
S1(config)#vlan 10
S1(config-vlan)#name USERS
S1(config-vlan)#exit
S1(config)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#exit
S1(config)#vlan 99
S1(config-vlan)#name MANAGEMENT
S1(config-vlan)#exit
S1(config)#vlan 999
S1(config-vlan)#name BLACKHOLE
S1(config-vlan)#exit
S1(config)#
```

ตั้งค่าชื่ออุปกรณ์และเส้นทางของเครือข่าย

Step 3: Secure Unused Ports (Security First)

```
None
S1(config)# interface range fa0/4-24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)# exit
```

```
S1(config)#interface range fa0/4-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport access vlan 999
S1(config-if-range)#shutdown
```

```
S1(config-if-range)#exit
```

Step 4: Assign Access Ports

None

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#
S1(config)#interface fa0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#
S1(config)#interface fa0/2
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 20
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#
S1(config)#interface fa0/3
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 5: Configure Trunk to Router

```
None
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)# exit
```

```
S1(config)#
S1(config)#interface g0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan 10,20,99
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
```

Step 6: Configure Management SVI

```
None
S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)# exit

S1(config)# ip default-gateway 192.168.10.129
S1(config)# end
S1# copy running-config startup-config
```

```
S1(config-if)#
S1(config-if)#
S1(config-if)#interface vlan99
S1(config-if)#ip address 192.168.10.131 255.255.255.192
S1(config-if)#no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#
S1(config)#ip default-gateway 192.168.10.129
S1(config)#
S1(config)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#
```

PART 2: ROUTER CONFIGURATION (30 minutes)

Step 7: Basic Router Setup

```
None
Router> enable
Router# configure terminal
Router(config)# hostname R1
```

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
```

Step 8: Enable Physical Interface

```
None
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config-if)# exit
```

```
R1(config)#interface g0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

Step 9: Configure Router-on-a-Stick Subinterfaces

```
None
R1(config)# interface g0/0.10
R1(config-if)# encapsulation dot1Q 10
R1(config-if)# ip address 192.168.10.1 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.20
R1(config-if)# encapsulation dot1Q 20
R1(config-if)# ip address 192.168.10.65 255.255.255.192
R1(config-if)# exit

R1(config)# interface g0/0.99
R1(config-if)# encapsulation dot1Q 99
```

```
R1(config-if)# ip address 192.168.10.129 255.255.255.192
R1(config-if)# exit
```

```
None
R1(config)# end
R1# copy running-config startup-config
```

```
R1(config)#
R1(config)#interface g0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config)#
R1(config)#
R1(config)#interface g0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.10.65 255.255.255.192
R1(config-subif)#exit
R1(config)#
```

```
R1(config)#
R1(config)#interface g0/0.99
R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.10.129 255.255.255.192
R1(config-subif)#exit
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

PART 3: END DEVICE CONFIGURATION (15 minutes)

Configure each PC manually.

PC-A (VLAN 10)

- IP: 192.168.10.10
- Mask: 255.255.255.192
- Gateway: 192.168.10.1

PC-B (VLAN 20)

- IP: 192.168.10.70
- Mask: 255.255.255.192
- Gateway: 192.168.10.65

PC-C (VLAN 99)

- IP: 192.168.10.130
- Mask: 255.255.255.192
- Gateway: 192.168.10.129

PART 4: CONNECTIVITY VERIFICATION (20 minutes)

Step 10: Verification Commands

From S1

```
None
show vlan brief
show interfaces trunk
```

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

From R1

```
None
show ip interface brief
```

```
R1#
R1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0     unassigned      YES unset  up
up
GigabitEthernet0/0.10  192.168.10.1    YES manual up
up
GigabitEthernet0/0.20  192.168.10.65   YES manual up
up
GigabitEthernet0/0.99  192.168.10.129  YES manual up
up
GigabitEthernet0/1     unassigned      YES unset  administratively down
down
GigabitEthernet0/2     unassigned      YES unset  administratively down
down
Vlan1                  unassigned      YES unset  administratively down
down
R1#
```

From PCs

```
None
ping 192.168.10.1
ping 192.168.10.70
ping 192.168.10.131
```

PC-A

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC-B

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time=5ms TTL=128
Reply from 192.168.10.70: bytes=32 time<1ms TTL=128
Reply from 192.168.10.70: bytes=32 time=3ms TTL=128
Reply from 192.168.10.70: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 2ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254
Reply from 192.168.10.131: bytes=32 time=2ms TTL=254
Reply from 192.168.10.131: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

PC-C

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.131

Pinging 192.168.10.131 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255
Reply from 192.168.10.131: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.131:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Expected:

- PC-A ↔ PC-B ✅
- PC-A ↔ S1 Management IP ✅

PART 5: TROUBLESHOOTING CHALLENGE (25 minutes)

Scenario A: Wrong VLAN Assignment

- Move PC-B to the wrong VLAN
- Test connectivity
- Identify failure point
- Restore correct VLAN

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk
- Test access to S1
- Explain why management fails
- Restore configuration

Scenario C: Wrong Gateway

- Change PC-A gateway
- Observe ping failure
- Use `ipconfig` and `ping`
- Fix gateway

📝 LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	Found 10, 20, 99, 999	Pass
Trunk active	show int trunk	VLANs allowed	10, 20, 99 allowed	Pass
Router subifs	show ip int br	up/up	up up	Pass

Checkpoint	Command	Expected	Actual	Pass/Fail
Inter-VLAN ping	ping	Success	Ping ติด 192.168.10.70	Pass

Worksheet 2: Design Understanding

Answer briefly:

1. Why is VLAN 99 used for management?
   Ans เพราะ vlan 99 ถูกใช้เพื่อสร้างช่องทางบริหารจัดการที่ปลอดภัยและเป็นอิสระ จากเครือข่าย

2. Why is VLAN 999 assigned to unused ports?
   Ans เพื่อป้องกันการโจมตีจากภายนอก และถ้าเกิดการโจมตีจะเปลี่ยนเส้นทางไปยัง vlan 999 ทันทีซึ่งมันถูกตัดขาดจาก LAN

3. What does 802.1Q tagging do?
   Ans 802.10 คือมาตรฐานสากลที่ใช้ในการทำ VLAN Tagging บนสายเชื่อมต่อที่เป็น Trunk หน้าที่หลักของมันคือ:
   ระบุตัวตนของข้อมูล: เมื่อข้อมูล (Frame) วิ่งออกจาก Switch ผ่านสาย Trunk ไปยัง Router หรือ Switch ตัวอื่น 802.1Q จะทำการแปะ "ป้ายชื่อ" หรือ VLAN ID ลงไปใน Header ของข้อมูลนั้น
   แยกแยะความเป็นเจ้าของ: เพื่อให้เครื่องรับ (เช่น Router R1) รู้ว่าข้อมูลที่ได้รับมานั้นมาจาก VLAN ไหน (เช่น VLAN 10 หรือ 20) และควรจะส่งต่อไปที่ไหนต่อ
   คงความเป็นส่วนตัว: ช่วยให้ข้อมูลของแต่ละ VLAN ไม่ปะปนกันแม้จะวิ่งอยู่บนสายเส้นเดียวกัน

4. Why does each VLAN require a different subnet?
   Ans เหตุผลที่ต้องแยก Subnet (เช่น 192.168.10.0/26 และ 192.168.10.64/26) สำหรับแต่ละ VLAN มีดังนี้:
   เพื่อให้ Router สามารถทำ Routing ได้: Router จะส่งข้อมูลข้ามเครือข่ายได้ก็ต่อเมื่อ "ต้นทาง" และ "ปลายทาง" อยู่คนละ Network ID กัน หากใช้ Subnet เดียวกัน Router จะมองว่าเป็นวงเดียวกันและไม่ยอมส่งข้อมูลข้ามไปให้
   การจำกัด Broadcast Domain: จุดประสงค์ของการทำ VLAN คือการแบ่งกลุ่มคอมพิวเตอร์เพื่อลดการรบกวนจาก Broadcast traffic หากใช้ Subnet เดียวกัน ประโยชน์ในการแยกกลุ่มก็จะหายไป
   ความปลอดภัยและการจัดการ: การแยก Subnet ช่วยให้เราสามารถตั้งค่า Firewall หรือ Access Control List (ACL) เพื่อเลือกว่าจะให้ VLAN ไหนคุยกับใครได้บ้างในระดับ Layer 3

🔍 VERIFICATION COMMANDS CHECKLIST

Router

- ☑ ~~show run~~
- ☑ ~~show ip route~~
- ☑ ~~show arp~~

```
R1#show run
Building configuration...

Current configuration : 979 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!

R1#
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.10.64/26 is directly connected, GigabitEthernet0/0.20
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0.20
C       192.168.10.128/26 is directly connected, GigabitEthernet0/0.99
L       192.168.10.129/32 is directly connected, GigabitEthernet0/0.99

R1#
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.10.10           15  00E0.F7ED.3856  ARPA   GigabitEthernet0/0.10
Internet  192.168.10.70           15  0001.6447.C798  ARPA   GigabitEthernet0/0.20
Internet  192.168.10.130          10  00D0.FFD5.9BB8  ARPA   GigabitEthernet0/0.99
Internet  192.168.10.131          15  0001.C7CE.A801  ARPA   GigabitEthernet0/0.99
R1#
```

Switch

- ☑ ~~show vlan brief~~
- ☑ ~~show interfaces trunk~~
- ☑ ~~show mac address-table~~

```
S1#
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

```
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99

S1#
S1#
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.a3a3.e501    DYNAMIC     Gig0/1
  10    0030.a3a3.e501    DYNAMIC     Gig0/1
  20    0030.a3a3.e501    DYNAMIC     Gig0/1
  99    0030.a3a3.e501    DYNAMIC     Gig0/1
S1#
```

PCs

- [x] ~~ipconfig~~
- [x] ~~arp -a~~
- [x] ~~ping~~

PC-A

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F7FF:FEED:3856
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.1          0030.a3a3.e501        dynamic
```

PC-B

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:64FF:FE47:C798
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.70
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.65

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.65         0030.a3a3.e501        dynamic
```

PC-C

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:FFFF:FED5:9BB8
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.130
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.129

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.10.129        0030.a3a3.e501        dynamic
  192.168.10.131        0001.c7ce.a801        dynamic
```

🎓 ASSESSMENT QUESTIONS

1. Which device performs inter-VLAN routing?
   Ans Router (R1) โดยPC ต่าง VLAN สามารถสื่อสารกันได้ เพราะ Router ทำหน้าที่เป็น Default Gateway คอยรับส่งข้อมูลข้ามระหว่าง VLAN ผ่านทาง Sub-interfaces บนสาย Trunk Link ทำให้เครือข่ายที่ถูกแยกกันทางตรรกะ (Logically Isolated) สามารถแลกเปลี่ยนข้อมูลกันได้

2. Why can PCs in different VLANs communicate?
   Ans เพราะ PC ส่งข้อมูลไปหา Router (R1) ซึ่งทำหน้าที่เป็น Default Gateway Router จะรับข้อมูลจาก VLAN แล้วส่งต่อ Route ไปยัง VLAN ปลายทางให้ ผ่านทางสาย Trunk

3. What breaks if the trunk is removed?
   Ans การสื่อสารระหว่าง VLAN จะตัดขาดทันที และจะไม่สามารถ Remote เข้าไปจัดการ Switch S1 ผ่าน IP Management ได้ เนื่องจากไม่มีช่องทางส่งข้อมูลไปยัง Router

4. Why is VLAN 1 avoided?
   Ans เพราะ VLAN 1 เป็นค่าเริ่มต้น และมักเป็นเป้าหมายการโจมตีทางเครือข่าย การย้ายทราฟฟิกไป VLAN อื่นที่กำหนดเองจึงปลอดภัยกว่า

5. What security risk exists if unused ports remain active?
   Ans ผู้ประสงค์ร้ายหรือคนนอกสามารถนำสาย LAN มาเสียบเข้ากับพอร์ตนั้น ผู้บุกรุกอาจทำการดักจับข้อมูล, ปล่อยไวรัส, หรือโจมตีแบบ MAC Flooding เพื่อทำให้ Switch ทำงานหนักจนระบบล่ม

🎯 LAB COMPLETION CRITERIA

✔ All VLANs correctly created ✔ Router-on-a-Stick functioning ✔ Management VLAN isolated ✔ Unused ports secured ✔ All troubleshooting scenarios resolved

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